

Trasformazioni del nostro algoritmo zk:

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 $x = (pk, a, id, tr, now), w = (MSO, pk_{dx}, pk_{dy})$   
 $e1 = SHA256(MSO[0 : 183])$   
 $a = MSO[id]$   
 $(pk_{dx}, pk_{dy}) = MSO[96 : 160]$   
 $tstart = MSO[48 : 56]$   
 $tend = MSO[56 : 64]$   
 $tstart < tnow < tend$   
 $true = p256.verify((r1, s1), e1, PKII)$   
 $true = p256.verify((r2, s2), H(tr || hdr), (pk_{dx}, pk_{dy}))$ 
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Post quantum transition

Issuer's signature proof zk-dillithium (STARK):

$x = (PK_{II}, C), w = (MSO, r)$

$C = SHA256(MSO, r)$

$e1 = SHA256(MSO[0 : 183])$

$true = Dillithium.verify((r1, s1), e1, PK_{II})$

Attribute disclosure, document validity and device binding proof (LIGERO):

$x = (a, id, tr, now, C), w = (MSO, pk_{dx}, pk_{dy}, r)$

$C = SHA256(MSO, r)$

$a = MSO[id]$

$(pk_{dx}, pk_{dy}) = MSO[96 : 160]$

$tstart = MSO[48 : 56]$

$tend = MSO[56 : 64]$

$tstart < tnow < tend$

$true = p256.verify((r2, s2), H(tr||hdr), (pk_{dx}, pk_{dy}))$

Verifier additional check: $STARK(C) = LIGERO(C)$

Senza mostrare all'issuer la device public key

Issuer's signature proof zk-dillithium (STARK):

$$x = (PK_{II}, C), w = (MSO, r)$$
$$C = \text{SHA256}(MSO, r)$$
$$e1 = \text{SHA256}(MSO[0 : 183])$$
$$true = \text{Dillithium.verify}((r1, s1), e1, PK_{II})$$

Attribute disclosure, document validity and device binding proof (LIGERO):

$$x = (a, id, tr, now, C), w = (MSO, pk_{dx}, pk_{dy}, r)$$
$$C = \text{SHA256}(MSO, r)$$
$$a = MSO[id]$$
$$\text{SHA256}(pk_{dx}, pk_{dy}) = MSO[96 : 160]$$
$$tstart = MSO[48 : 56]$$
$$tend = MSO[56 : 64]$$
$$tstart < tnow < tend$$
$$true = p256.verify((r2, s2), H(tr || hdr), (pk_{dx}, pk_{dy}))$$

Ottimizzazione del commitment

Il nuovo commitment è $C = H(MSO) \oplus r$. Eliminiamo di fatto una prova su SHA in zkSTARK.

Issuer's signature proof zk-dillithium (STARK):

$x = (PK_{II}, C)$, $w = (MSO, r)$

$e1 = \text{Poseidon}(MSO[0 : 183])$

$C = e1 \oplus r$

$true = \text{Dillithium.verify}((r1, s1), e1, PK_{II})$

Attribute disclosure, document validity and device binding proof (LIGERO):

$x = (a, id, tr, now, C)$, $w = (MSO, pk_{dx}, pk_{dy}, r)$

$C = \text{Poseidon}(MSO) \oplus r$

$a = MSO[id]$

$SHA256(pk_{dx}, pk_{dy}) = MSO[96 : 160]$

$tstart = MSO[48 : 56]$

$tend = MSO[56 : 64]$

$tstart < tnow < tend$

$true = p256.verify((r2, s2), H(tr || hdr), (pk_{dx}, pk_{dy}))$